Riya Bansal

Embedded Firmware Engineer

RESUME_1249

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Total Experience: 5 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 5 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: ARM Cortex-M, Embedded Linux, FreeRTOS, STM32, C, I2C, C++, UART, SPI, JTAG, CAN, Git

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | WillowGrid Labs | Jan 2023 - Present

- Lowered field-reset incidents by 12% with watchdog recovery and diagnostics
- Improved battery runtime by 39% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 25% using interrupt priority and queue tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | ZenithTrail Technologies | Jul 2021 - Dec 2022

- Reduced embedded task jitter by 19% using interrupt priority and queue tuning
- Validated 476 hardware test vectors across protocol and timing corner cases

PROJECTS

Motor Control Interface | I2C, C++, STM32, FreeRTOS

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Low-Power Metering Device | CAN, STM32, FreeRTOS, ARM Cortex-M

- Optimized interrupt-driven acquisition and sleep cycles.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2021
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2024)